

Ayman Charania

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EDUCATION

- Rice University** Houston, TX
 - BA in Computer Science; GPA: 4.0 (President's Honor Roll)* *Aug. 2022 – May 2025*
 - Relevant Coursework:* Computational & Algorithmic Thinking, Computer Engineering, Linear Algebra, Functional Programming, Concurrent Program Design, Operating Systems, Data Science, Production Programming, Compiler Construction

EXPERIENCE

- Software Engineering Intern** Beaumont, TX
 - Xplor Spot Munro's DCC — JavaScript, HTML, CSS, SQL, Node.js* *May 2024 - August 2024*
 - Engineered backend features that streamlined order processing and payment integration, improving throughput and cutting service wait times by 30%.
 - Aligned system improvements with stakeholder priorities by translating business needs into technical specs, resulting in smoother customer interactions and fewer support tickets.
 - Designed automated and manual test suites to catch edge cases early, reducing post-deploy bugs by 40% and supporting 99.9% uptime.
- Software Engineer** Houston, TX
 - ACT@Rice (Recipe for Success App) — React Native, Node.js, WordPress* *August 2023 - May 2024*
 - Led the development of a nonprofit mobile app promoting healthy eating, directly impacting 500+ users through an engaging, gamified UI.
 - Built a challenge tracker with a live leaderboard, boosting user participation by 60% through real-time feedback and progress tracking.
 - Architected a scalable frontend integrated with a WordPress backend, ensuring fast load times and low-latency data syncing.
- Front-End Software Engineering Intern** Beaumont, TX
 - CRDN — JavaScript, HTML, CSS, Node.js* *May 2023 - August 2023*
 - Developed backend systems for a disaster recovery service portal, enabling 24/7 handling of textile, art, and electronics restoration requests.
 - Automated customer request workflows via dynamic form logic, reducing intake processing time by 50% and improving accuracy of service orders.
- Data Intern** Beaumont, TX
 - Sunshine Stores Inc. — SQL, Excel, Python* *March 2022 - August 2022*
 - Performed deep-dive analysis across 30+ retail locations, uncovering performance trends that informed inventory and staffing decisions.
 - Executed a full data migration with integrity checks, boosting system reliability and supporting a successful shift to a new management platform.

PROJECTS

- M3ssag1n8 Web Client (Full Stack):** Built a TypeScript-based messaging client using OwlDB with env-based config, structured logging, hot reload, and E2E testing. *TypeScript, CSS, HTML* (Dec '24)
- Employee Management System (Full Stack):** Built a real-time admin dashboard for Munro's Cleaners to manage employee roles and payroll, cutting manual processing time by 50%. *JavaScript, CSS, HTML* (Aug '24)
- OwlDB - NoSQL Document Database (Full Stack):** Engineered a custom NoSQL database in Go with CLI support and concurrent skip list indexing, handling thousands of document operations with low latency. *Golang* (Aug '24)
- Bakery Website (Full Stack):** Developed a responsive, SEO-optimized site to showcase desserts and drive local customer engagement for a Houston-based bakery. *HTML, CSS, JavaScript* (May '24)
- Hurricane Harvey Effect on Crime Rates (Data Science):** Led data analysis on post-Harvey crime trends using Python and ML, earning 2nd place at Rice's DEEP competition for actionable insights. *Python, scikit-learn, numpy, pandas* (Dec '23)
- Spot It Game (Projective Geometry):** Created an interactive "Spot It" card game using projective geometry to ensure unique combinations and enhance educational gameplay. *Python* (Dec '23)
- Stock Prediction (Markov Chains):** Built a Markov Chain-based model to forecast stock prices and evaluate prediction accuracy against baseline strategies *Python* (Oct '23)
- Map Search Algorithms (Algorithm Design):** Implemented and visualized BFS, DFS, and A* algorithms to compare pathfinding efficiency in real-world street maps. *Python* (July '23)

PROGRAMMING SKILLS

- Languages:** Python, Java, JavaScript, CSS, HTML, C, Golang, Racket, TypeScript, SQL
- Technologies:** React, Pandas, Git, node.js, MySQL, Excel, PowerPoint, Word, LaTeX